MILITARY SPECIFICATION

SHIELDED AND CONFINED MILD DETONATING CORD ASSEMBLIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 General. This specification covers the general requirements for the design, development and preproduction testing of Shielded and Confined Mild Detonating Cord Assemblies to be utilized for aircrew escape systems. Shielded Mild Detonating Cord and Flexible Confined Detonating Cord hereafter referred to SMDC and FCDC respectively.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-200	Aluminum Alloy, Bar, Rod, Shapes, Tube and Wire, Extruded and Structural Shapes, General Specification for
QQ-A-225	Aluminum and Aluminum Alloy Bar, Rod, Wire or Special Shapes; Rolled Drawn or Cold Finished, General Specification for
FED-STD-595	Colors

Military

MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-C-5541	Chemical Film and Chemical Film Material for Aluminum and Aluminum Alloys
MIL-I-6866	Inspection, Penetrant, Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice Process for)
MIL-F-7179	Finishes and Coatings: General Specification for Protection of Aerospace Weapons Systems, Structrues and Parts
MIL-S-7742	Screw Threads, Standard Optimum Selected Series, General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-18303	Test Procedures; Preproduction Acceptance and Life for Aircraft Electronic Equipment, Format for

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MIL-P-22808	Paint, Epoxy, Hydraulic Fluid Resistant
MIL-P-23377	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-A-25165	Aircraft Emergency Escape System Identification of
MIL-U-81055	Ultrasonic Inspection
MIL-L-81352	Lacquer: Acrylic (for Naval Weapons)

STANDARDS

Military

MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-331	Fuze and Fuze Components, Environmental and performance tests for
MIL-STD-453	Inspection, Radiographic
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program
MIL-STD-889	Dissimilar Metals
MIL-STD-1166	Radiographic Testing Requirements for Solid Propellant
MIL-STD-1316	Fuses, Navy, Design Criteria for
MS17983	Compass, Magnetic, Pilot's Standby
MS20995	Wire, Lock
MS21919	Clamp, Cushioned, Support, Loop-Type Aircraft ASG
MS33540	Safety Wiring, General Practices for

HANDBOOKS

Military

MIL-HDBK-5	Metallic Materials and Elements for Aerospace Vehicle Structures
ANA-438	Age Controls of Age Sensitive Elastomeric Items

Navy

WR-62	Naval Weapons Requirements, Specifications and Standards: Use of
WS 4660	Dipam Explosive(U) Purchase Description
WS-5003	HNS Explosive (U) Purchase Description

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3. REQUIREMENTS

3.1 Preproduction.- This specification makes provision for preproduction testing.

3.2 Selection of Specification and Standards.- Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.- All materials used in propellant formulation, hardware components, adhesives etc., should be in accordance with existing government specifications. Use of materials not covered by existing Government specifications shall require prior approval by the cognizant design agency.

3.3.1 Metals.- Metals shall be of the corrosion resisting type or shall be suitably protected to resist corrosion in all the environments required by this specification, when in storage or during normal service life. Unless otherwise specified herein, physical properties of all metals shall meet the minimum requirements specified in MIL-HDBK-5.

3.3.1.1 Dissimilar Metals.- Dissimilar metals, as defined in MIL-STD-889, shall not be used in contact with each other. Where such contact is unavoidable, the parts shall be protected in accordance with MIL-F-7179.

3.3.1.2 Heat treatment.- Heat treatment of steel parts shall be in accordance with Specification MIL-H-6875.

3.3.1.3 Aluminum.- Where a high strength aluminum alloy is required, 7075-T73 2024T3 or 6061T6 in accordance with Specification QQ-A-225 (wrought) and QQ-A-200 (extruded) shall be used.

3.3.1.4 Magnesium and Magnesium Alloys.- Magnesium and magnesium alloys shall not be used.

3.3.1.5 Castings.- When used, castings for structural applications require procuring activity approval prior to use.

3.3.2 Limited life items.- The use of materials and elements whose life is anticipated to be less than the required life of the assemblies shall not be used.

3.3.3 Reaction of materials.- Materials shall not be used which may produce and emit toxic, nauseating, or corrosive products under the environmental conditions specified herein. In addition, materials shall be capable of functioning in the fluids, vapors or gases specified herein, before, during and after exposure to all the environmental conditions specified herein.

3.3.4 Non-metals.- When plastics, elastomeric and nonmetallic materials other than military approved articles are used, adequate proof of their suita-

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Cover Assembly

bility shall be submitted along with test articles submitted for evaluation testing. The use of these articles shall be with proper consideration of their resistance to air and hydraulic fluid (where applicable) including the special problems of dimensional stability, aging characteristics, moisture absorption, and thermal embrittlement.

Liquids shall not be used in final assembly unless approved by the procuring activity. Age controls for elastomers shall be in accordance with ANA Bulletin No. 438.

3.3.5 Fungus-proof materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.3.6 Parts.

3.3.6.1 Inspection of metal parts.- All highly stressed steel parts shall be subjected to magnetic particle inspection in accordance with Specification MIL-I-6868 or a similar method. Titanium, aluminum or nonmagnetic steel high stress parts shall be inspected in accordance with Specification MIL-I-6866 penetrant inspection.

3.3.7 Protective treatment.- All materials used which are not inherently corrosive resistant, shall be treated in accordance with Specification MIL-F-7179 to resist any corrosive effects resulting from the environmental conditions specified herein. Any protective coatings used shall not crack, chip, erode, scale with age or degrade the structural integrity of the materials used in construction of the assemblies when exposed to conditions specified herein.

3.3.7.1 Aluminum.- All aluminum and aluminum alloys shall be anodized in accordance with Specification MIL-A-8625 Type II coatings except that in the absence of abrasive conditions, the anodize may be Type I coating where applicable, except for MDC sheath. Repair or touch up shall be by application of chemical film in accordance with Specification MIL-C-5541, Grade B.

3.3.7.2 Priming and painting.- In addition to the surface treatments herein, all external surfaces of the assemblies shall be primed and painted as follows, except where such finish would impair proper function.

(a) One coat of epoxy primer conforming to Specification MIL-P-23377 or MIL-P-22308.

(b) One acrylic or epoxy topcoat conforming to Specification MIL-L-81352.

(c) Color in accordance with Specification FED-STD-595 and subject to procuring activity approval.

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3.4 Design and construction.- The configuration of the SMDC and the inert connectors shall be as shown in Figure 3.

3.4.1 Weight (Guaranteed).- The guaranteed weight of the assemblies and the inert connectors shall be kept to a minimum consistent with the requirements of this specification.

3.4.2 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable as defined in Specification MIL-I-8500. The drawing number requirements of MIL-D-1000 shall govern the changes in manufacturer's part numbers.

3.4.3 Workmanship.- Workmanship shall be in accordance with high grade aircraft practice and of a quality to assure safety, proper operation, and service life. Workmanship shall be inspected and certified by the contractor quality control.

3.4.4 Radiographic Inspection.

3.4.4.1 X-Ray inspection.- The assemblies shall be X-Ray inspected along their straight length and ends only after forming, in accordance with Standard MIL-STD-453. X-Ray film shall be retained. The accept/reject criteria to be used in the evaluation of the X-rays shall be coordinated between the contractor and the procuring activity and shall be approved prior to qualification.

3.4.5 Residual magnetism.- All parts made of materials that are capable of retaining residual magnetism shall be demagnitized sufficiently to prevent component malfunction due to accumulation of magnetic contaminants.

3.4.6 Installation requirements.- The production assembly configurations shall correspond to the inert masters furnished by the procuring activity. All SMDC assemblies shall meet the following requirements.

3.4.6.1 SMDC & FCDC end fittings.- The SMDC assemblies when installed in the ports shall be capable of withstanding A70-T090 IN-LBS torque values:

3/8 - 24 UNF - 3A THD, 1/2 - 20 UNF-3A THD

7/16 - 20 UNF - 3A THD, 5/8 - 18 UNF - 3A THD

FCDC End Fittings.- The FCDC assemblies when installed in the port configuration as shown in Figures 1 and 1A shall be torqued as installed prior to firing. The fitting shall be capable of withstanding 90 inch-pounds of torque. All FCDCs shall remain intact in port areas after firing. The end fitting joint of the FCDC when mounted into fittings indicated in Figure 3 shall be capable of withstanding a tensile load of 350 pounds minimum applied at a rate of 8 + 2 pounds per second.

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3.4.6.2 Stainless steel tube segments.- The stainless steel tube segments of the SMDC after assembly shall be capable of being formed to a 3/4 inch radius helix on the centerline of the assembly without rupturing the tubing or detrimentally affecting the performance requirements specified herein.

3.4.6.3 Flexibility.

3.4.6.3.1 Final assemblies.- The final assemblies of the FCDC shall be capable of being formed around a 2.50 inch diameter mandril without rupturing the plastic covering or detrimentally affecting the performance requirements specified herein.

3.4.6.3.2 The final assembly FCDC.- Units shall be capable of 20,000 flexure cycles of 40° in one plane. With the FCDC assembly in a vertical position and held firmly at one end on a surface, flex the unit 20° in one direction; back to vertical then 20° in the opposite direction and back to vertical for one (1) complete cycle.

3.4.7 Sealing requirements.- The assemblies shall be sealed to prevent penetration by water, moisture, and dust with a maximum leakage rate of 1×10^{-5} std cc/sec of air at a pressure differential of $\pm$ 0.1 atmosphere. All gases produced by actuation shall be contained to preclude deleterious effects on aircraft systems and occupants.

3.4.8 Threads.- All threads shall be in accordance with Specification MIL-S-7742.

3.4.9 End fitting strength.- The end fitting joint between the outer ferrule and the stainless steel tubing shall be capable of withstanding a tensile load of 1000 lbs. minimum applied at a rate of 8 $\pm$ 2 pounds per second. The specimen need not function after the test.

3.4.10 Structures.

3.4.10.1 SMDC structural integrity.- The SMDC assembly shall be constructed such that the portion of the assembly, external of the SMDC port, remains intact without rupturing, splitting, cracking or disengaging from the end fittings and without emitting shrapnel when fired in the "as installed" condition at +350°F after being stabilized at this temperature for one hour. The SMDC assembly is required to meet the aforementioned when fired with the end tips or boosters confined in blind ports having a free volume of not greater than 0.03 cubic inches.

3.4.10.2 FCDC structural integrity.- The FCDC assembly shall be constructed such that the portion of the assembly protruding from the installation ports shall remain intact without rupturing, cracking, splitting or disengaging from the end fittings. The FCDC is required to meet the above when fired with the end tips or boosters confined in blind ports having a volume of 0.03 cubic inch, at +200°F.

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3.4.11 Detonating cord charge.- The detonating cord (DC) charge shall be the chemical compound Dipicramide (DIPAM Grade A) in accordance with WS4660 or chemical compound hexanitrostilbene (HNS-II) Grade A in accordance with WS5003 and shall provide sufficient output to continue the explosive train and detonate the booster caps. The minimum detonating velocity, moisture content, and core loading tolerance of the DC shall be determined by the contractor and acceptable values agreed upon by the procuring activity and contractor prior to the beginning of preproduction testing. The DC charge shall meet the sensitivity requirements of MIL-STD-1316 and the performance and reliability requirements specified herein.

3.4.12 Explosive transfer.- The assemblies shall be capable of propagating detonation when fired in the test fitting configuration as stated in Par (a). The contractor shall submit details of the design which define the following:

(a) Maximum allowable misalignment between two SMDC end assemblies caused by tolerances on the SMDC design for each firjng configuration of the test fitting shown in Figure 3.

Maximum misalignment permissible between two (2) FCDC end assemblies caused by tolerance build up on the FCDC for each firing configuration of the test fittings shown on Figures 5 and 6.

(b) The amount of the explosive and the tolerance for the SMDC and FCDC end booster design broken down by each component which has an explosive charge.

(c) SMDC and FCDC end booster cup material, thicknesses and tolerances.

3.4.13 SMDC and FCDC end booster.- The explosive material of the SMDC and FCDC end booster and transition charge between the booster and the MDC shall be the chemical compound hexanitrostilbene (HNS-I) Grade A in accordance with Publication WS5003, and shall meet the sensitivity requirements of Standard MIL-STD-1316. The end booster shall contain 1 $\pm$ .1 grain of HNS pressed to 32,000 $\pm$ 1,000 PSI. Any "pick-up" charges between the detonating cord and end booster shall also be HNS-1 Grade A.

3.4.14 Fastening.- The SMDC and FCDC end fitting shall have provisions for lockwiring to a mating device in accordance with Standard MS20995 and MS33540, or utilizing a standard self locking nut.

3.4.15 Standard SMDC test assembly.- The Standard SMDC Test Assembly as shown in Figure 6 shall be totally representative of a typical production SMDC line and shall conform to all parameters of this specification.

3.4.16 Explosive cord sheath.- The SMDC explosive cord shall consist of an explosive core inclosed in an aluminum sheath and the FCDC explosive cord shall consist of an explosive core inclosed in a silver sheath.

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3.4.17 Environmental conditions.

NOTE: For all dynamic environments the SMDC shall be mounted as shown in Figure 7.

3.4.17.1 Standard conditions.- The following conditions shall be used to establish normal performance characteristics and for making laboratory bench tests. (The conditions stated are for the device being tested).

Temperature	70° ± 5°F
Altitude	Normal Ground
Vibration	None
Humidity	95% maximum

3.4.17.2 Service condition.- The SMDC and FCDC shall meet all the requirements of this specification during or after exposure to the environmental service conditions or combination of these conditions specified herein.

3.4.17.2.1 Temperature.- The SMDC and FCDC shall meet all the performance requirements of this specification during and after the required exposure periods to the following temperatures:

Service	Minus 65°F to plus 350°F (SMDC)
Service	Minus 65°F to plus 200°F (FCDC)
Storage (both)	Minus 80°F to plus 160°F

The SMDC shall also be capable of withstanding temperatures of plus 400°F for periods of five minutes repeatedly.

3.4.17.2.2 Drop.

3.4.17.2.2.1 Forty-foot drop.- The assemblies shall not initiate and shall be safe to handle and dispose of when dropped from a height of 40 feet. The assemblies are not required to operate after being exposed to the test specified in 4.5.1.4.

3.4.17.2.2.2 Six-foot drop.- The assemblies shall not detonate and shall be capable of satisfactory operation after being subjected to the test specified in 4.5.1.5.

3.4.17.2.3 High temperature.- The SMDC and FCDC assemblies shall be capable of withstanding the temperatures as specified in 4.5.1.2.8 and shall meet the design performance requirements when test fired.

3.4.17.2.4 Auto detonation.- This is a safety test designed to assure that the system will not be initiated by thermal exposure. This test shall be conducted as specified in para. 4.5.1.2.9.

3.4.17.2.5 High temperature exposure (12 hours).- The maximum temperature (within 25°F) shall be established to which SMDC and FCDC may be exposed for 12 hours and function within design performance requirements when fired

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3.4.17.2.6 DELETED

after cooling to 70°F. The maximum temperature shall be determined by the high temperature exposure test as specified in 4.5.1.2.10.

3.4.17.2.6 Temperature/shock/humidity/altitude.- The assemblies shall be capable of withstanding temperature shock/humidity/altitude cycling as specified in 4.5.1.2.7 and shall subsequently meet the design performance requirements when test fired.

3.4.17.2.7 Salt-fog.- The assemblies shall be capable of withstanding exposure to salt-fog conditions as specified in 4.5.1.2.2 and shall subsequently meet the design performance requirements when test fired.

3.4.17.2.8 Vibration.- The assemblies shall not fire when subjected to vibration as specified in 4.5.1.2.4. After subjection to such vibration, the assemblies shall meet the design performance requirements when test fired.

3.4.17.2.9 Shock.- The assemblies shall be capable of withstanding shock conditions with peak amplitudes of 20 g for a duration of 11 ms as specified in 4.5.1.2.5 and shall subsequently meet the design performance requirements when test fired.

3.4.17.2.10 Dust.- The assemblies shall be capable of withstanding exposure to dust-laden atmosphere as specified in 4.5.1.2.3 and shall subsequently meet the design performance requirements when test fired.

3.4.17.2.11 Submerged firing.- The SMDC and FCDC assembly shall be capable of firing while immersed in water and at a pressure of 50 psig. as specified in 4.5.1.2.7.

3.4.18 Product marking.- Each assembly shall be clearly and permanently identified with the SMDC and/or FCDC designation, identifying number, lot number, loading date (month and year), manufacturer's identification symbol and in accordance with MIL-A-25165.

3.4.19 Foolproofness.- The assemblies shall be designed to prevent incorrect assembly and installation.

3.4.20 Ruggedness.- The assemblies shall withstand ordinary handling and manipulation by unskilled personnel throughout the course of shipping, inspection, installation and service.

3.4.21 Performance reliability.- The assemblies must meet all performance requirements at a reliability of not less than 99.9 percent at a confidence level of 90% over the temperature range of -65°F to +350°F for SMDC and -65° to +200°F for FCDC.

3.4.21.1 Inadvertent fire probability.- The inadvertent fire probability of the assemblies shall be zero under any environmental conditions specified herein. Any evidence of the assemblies to fire prematurely shall be cause for rejection of the design. An inadvertent firing is defined as an involuntary, or premature ignition of any explosive parts of the assembly.

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3.4.21.2 Life requirements.- The assembly shall have a total life eight years minimum including at least 5 years of service life. During this time, the assembly shall not suffer deterioration or loss of performance beyond the limits specified herein.

3.4.21.2.1 Storage life.- Storage life is defined as that period during which assemblies are stored in their original factory-sealed storage container, and begins with the date of manufacture. Under these conditions the temperature will not exceed the limits of -80°F to +160°F. Assemblies shall be capable of being stored in any altitude.

3.4.21.2.2 Service life.- Service life is defined as the period of time beginning when the assembly storage container is unsealed and opened, whether the assembly is installed or not. During this period the assembly may experience any combination of environmental conditions specified herein.

3.5 Performance.

3.5.1 General.- The assemblies shall be detonated when installed in conjunction with other similar devices and shall meet the energy output requirements specified in the detail design performance requirements. The assemblies shall transfer detonation from one assembly to another while placed in the relative positions and at the air gaps specified in the detail requirements. A margin of safety must be demonstrated by transferring detonations over air gaps having 50 per cent greater widths than those specified in the detail requirements.

3.6 Development testing.- Development tests are conducted to provide data to be used in the design or in support of the design of a specific component or subsystem. Development tests may be used to determine operating characteristics under off-design conditions. Stress to failure development tests shall be conducted to provide failure mode characteristics for verification of analysis and determination of strength and operating margin of safety. It is required that development tests be run on the SMDC and FCDC under simulated thermal environment to assure the design compatibility with the operating requirements of this specification. Development tests shall be categorized as follows:

(a) Design Feasibility Tests (3.6.1)
(b) Design Verification Tests (3.6.2)

3.6.1 Design feasibility tests (DFT).- Design feasibility tests shall include all test conducted for the following purposes:

(a) Component and part selection.

(b) Investigation of the performance of the SMDC's and FCDC's shall include as a minimum, performance or functional tests, to indicate rupture or fragmentation, firing and velocity, shock, vibration, humidity, operating margins, minimum charge (explosive), nominal flight charge and structural integrity.

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(c) Demonstration of safety and operating margins or other analytical assumptions.

(d) Design Feasibility Tests results will establish the criteria for acceptance/rejection limits of detrimental deformation.

3.6.2 Design Verification Tests (DVT).- Design Verification Tests are advanced stage development tests which are performed on the assemblies SMDC for the purpose of substantiating the design for its intended mission. These tests shall be conducted under simulated mission environments and off-design conditions and shall include, as a minimum: shock, vibration, humidity, exhaust gas verification tests, firing and velocity tests, structural integrity tests and X-ray inspection. Successful completion of design verification tests provide the assurance to permit a design freeze decision to be made. The test shall use typical SMDC's and FCDC's permanently identified and numbered for the purpose of design verification tests. The contractor shall have available for review by the procuring activity all detail, in-process and final inspection records for each design verification SMDC and FCDC.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, if approved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Materials.- Inspection and testing of component parts and assemblies shall be made to determine compliance with 3.3. Where defects, incompatibility, or inferior quality is evident, and the Government deems material analysis necessary, the contractor will be requested to submit samples or specimens to the cognizant design agency for analysis and approval.

4.2 Classification of tests.- The inspection and testing of SMDC and FCDC shall be classified as preproduction and special tests.

4.3 Test conditions.

4.3.1 Measurements and records.- All functional tests of SMDC and FCDC shall be performed with the equipment mounted and loaded in a manner to simulate service conditions as closely as possible.

4.3.2 Instrumentation.- Any state-of-the-art-type instrumentation and recording equipment may be used; e.g., oscillograph, oscilloscope, or magnetic tape. Transducers may be any state-of-the-art type; e.g., piezo-electric, strain gage, variable reluctance, capacitive, or potentiometer. However, accuracy

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of all test apparatus shall conform to the requirements of MIL-STD-810. Documented calibration records will be maintained and be available for inspection by the cognizant design agency.

4.4 Test Methods.

4.4.1 Component inspection.- Each part of the assembly shall be 100 percent inspected for conformance in the applicable drawing.

4.4.2 Examination of product.- Examination of the assemblies shall be made to detect visible defects and to determine that the assemblies meet the finish, and marking requirements and that they conform to the applicable drawing.

4.4.3 Leak detection.- All assemblies shall pass a dry gas leak test. Assemblies which exhibit a leak rate in excess of 10^{-5}cc per second of air at a pressure differential of 1 $\pm$ 0.1 atmosphere shall be considered defective.

4.4.4 Demagnetization test.- The assembly shall be checked for residual magnetism in an area free of local magnetic effects by placing the device 6 inches from a pilot's standby compass, type MS17983 or equivalent, in a north-sourh horizontal position with the compass placed in an east-west heading as shown in Figure 9. The device shall be placed at the same height from the floor as the compass in a longitudinal direction along its entire length. This procedure shall be repeated for every 90 degrees rotation of the device. The assembly shall not deflect the compass more than three degrees.

4.4.5 X-Ray inspection.- The assemblies shall be X-ray inspected along their straight length and ends only after forming, in accordance with Standard MIL-STD-453. The accept/reject criteria to be used in the evaluation of the X-ray shall be coordinated between the contractor and the procuring activity prior to commencement of production.

4.5 Preproduction tests.- Preproduction tests shall be made on equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.7.

NOTE: When preproduction tests are to be made on equipments manufactured in the first production run, the equipments to be tested shall be selected by a representative of the procuring activity with the concurrence of the Government Inspector.

4.5.1 Scope of preproduction tests.- The specimen configuration shall be defined in detail spec. Preproduction tests shall include all tests specified herein and shall be conducted in accordance with the requirements of this specification. Test values and parameters shall be as designated in the design requirements paragraphs of this specification. If during the course of conducting preproduction tests corrective action is required, the contractor

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shall re-evaluate the previously approved tests, test plans, and test procedures to assure their continued validity. The corrective action, extent of required retest, and the test plan and procedure revisions shall be subject to approval of the procuring activity. Preproduction tests shall be performed on 170 SMDC's or 155 FCDC's in the following sequence and quantities shown in Table I and II:

Tests	Paragraph Numbers
Functional Test	4.5.1.1
Environmental Tests	4.5.1.2
Minimum Bend	4.5.1.3
Forty-Foot Drop Test	4.5.1.4
Six-Foot Drop Test	4.5.1.5
Flexibility	4.5.1.6

4.5.1.1 Functional test.

4.5.1.1.1 Cord test.- A sample length of the cord shall be tested for detonation velocity, moisure, content, and core loading for each 100 feet of length manufactured. Acceptable values shall be as specified in 3.4.11. The cord source, the sample length and the sample location shall be specified and reported in the preproduction test report.

4.5.1.1.2 Functional test firing.- 81 SMDC's shall be test fired as follows:

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
9	3	3	3	4.5.1.1.3.1(a)
9	3	3	3	4.5.1.1.3.1(b)
9	3	3	3	4.5.1.1.3.2(a)
60 FCDC's shall be test fired as follows:				
11	5	3	3	4.5.1.1.3.1(c)
11	5	3	3	4.5.1.1.3.1(d)

4.5.1.1.3 Test firing.

4.5.1.1.3.1 Output and Sensitivity.

(a) Tee fitting (Inert Connector).- An SMDC assembly shall be fired as a donor through an orifice while aligned as shown in Figure 4. A second

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SMDC assembly acting as an acceptor shall be initiated by the particles of the donor which pass through the orifice. The output shall be verified by a minimum plate dent of 0.040 in a 1 x 1 x .5 inch thick 6061-T6 aluminum block when the unprotected block and the SMDC tip are in intimate contact.

(b) Tee fitting (Inert Connector).- An SMDC assembly shall be fired as a donor through an orifice while aligned as shown in Figure 5. A second SMDC assembly acting as an acceptor shall be initiated by the particles of the donor which pass through the orifice. The output shall be verified by a minimum plate dent of 0.040 in a 1 x 1 x .5 inch thick 6061-T6 aluminum block when the unprotected block and the SMDC tip are in intimate contact.

(c) Two FCDC's shall be fired while aligned as shown in Figure 4. The first FCDC shall be fired using a standard SMDC tip as a donor. The first FCDC shall act as a donor for the second FCDC. The second FCDC assembly acting as an acceptor shall be initiated by the particles of the donor which pass through the orifice. The second FCDC shall deliver a minimum amount of energy as evidenced by a minimum dent of 0.040 inches into a 6061 T6 aluminum block 1.0 x 1.0 x .500 inches thick when the unprotected block and FCDC tip are in intimate contact.

(d) The FCDC shall be fired using a standard SMDC tip as a donor while aligned as shown in Figure 6. The FCDC shall transfer detonation to a second standard SMDC aligned as shown in Figure 6.

4.5.1.1.3.2 Output.

(a) Production assemblies.- An assembly shall be fired, while aligned as shown in Figure 6. The output shall be verified by propagation of the acceptor tip.

4.5.1.1.3.3 Output (Tensile Test).- An assembly shall be fired as shown in Figure 10. The output shall be verified by propagation of the acceptor tip.

4.5.1.1.4 Temperature conditioning for firing.- When specified, in Tables the assemblies shall be placed into a thermal environmental chamber at atmospheric pressure and conditioned as follows:

(a) High temperature. The temperature of the SMDC's under test shall be raised to +350°F and the FCDC's to +200°F and maintained at this temperature for a minimum of one hour after the assembly temperatures have been stabilized. The assemblies shall be fired while stabilized at this temperature.

(b) Ambient temperature.- These tests shall be conducted at +70°F ± 5°F and the assemblies fired while stabilized at this temperature.

(c) Low temperature.- The SMDC and FCDC under test shall be cooled to -65°F and maintained for a minimum of one hour. After the assembly temp. stabilized the assemblies under test shall be fired while stabilized at this temperature.

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4.5.1.2 Environmental tests.- The following environmental tests shall be performed on the number of SMDC's specified in Table I or FCDC's specified in Table II together with the required amount of inert connectors to perform the firing function. The contractor may start the next environmental sequence while conducting firing tests of the previously withdrawn samples. In no case shall the second subsequent environmental test be started without prior approval of the cognizant design agency.

4.5.1.2.1 Temperature shock/humidity/altitude test.- The assemblies shall be subjected to the temperature shock/humidity/altitude test as outlined below. The schedule has been arranged in such a manner that operations are not required outside normal working hours except for such supervision as may be necessary to insure proper operation of the test equipment. It is not mandatory that the day/clock time schedule given below be followed: however, it is mandatory that the time, environmental, and sequence requirements be adhered to. It is also noted that only two conditioning chambers are required to accomplish this test. No less than two chambers may be used and still accomplish the temperature shock portion of this test. It is permissible to use three chambers provided the time, environmental, and sequence requirements are met. The fluctuations from the specified temperatures shall not exceed 5°F. When the temperature/altitude chamber door is opened to place test items inside, the chamber pressure will become atmospheric. The time required to return the chamber pressure to 0.65 pounds per square inch (psi) shall not exceed 1 hour. Test specimens shall be supported on screen trays or racks so that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test.

Procedure

Monday 0800 - Place test items in a chamber maintained at +70°F at 50 percent relative humidity (RH).

1200 - Raise chamber temperature to +160°F and the RH to 95 percent. The chamber temperature shall reach +160°F at 95 percent RH not later than 1300.

1600 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (0.65 psi)

Tuesday 0800 - Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH.

1200 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (0.65 psi)

1600 - Remove test items from above chamber and immediately place in a chamber maintained at +160°F at 95 percent RH.

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Wednesday 0800 - Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 0900.

1200 - Raise chamber temperature to +160°F at 95 percent RH. The chamber temperature shall reach +160°F at 95 percent RH not later than 1300.

1600 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (o.65 psi)

Thursday 0800 - Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH.

1200 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (0.65 psi)

1600 - Remove test items from above chamber and immediately place in a chamber maintained at +160°F at 95 percent RH.

Friday 0800 - Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 0900.

1200 - Raise chamber temperature to +160°F at 95 percent RH. The chamber temperature shall reach +160°F at 95 percent RH not later than 1300.

1600 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at standard ambient pressure.

This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +160°F at 95 percent RH. At the conclusion of the temperature-shock/humidity/altitude test, the test items shall be allocated to the tests specified in Tables I or II. Assemblies shall be fired as follows within four hours after removal from the environmental chamber.

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1(a)
2		2		4.5.1.1.3.1(b)
3	1	1	1	4.5.1.1.3.2(a)

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FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1 (c)
5	2	1	2	4.5.1.1.3.1 (d)

4.5.1.2.2 Salt-fog.- Prior to subjecting the assemblies to the salt-fog environment, six assemblies shall have end caps installed. These six assemblies and the inert connectors required for each test firing shall be subjected to the environments and subsequently fired during the submerged operation test specified in 4.5.1.2.7. The salt-fog test shall be conducted in accordance with Method 509 of MIL-STD-810. Upon completion of the salt-fog test, assemblies shall be fired as follows within four hours after removal from the salt-fog chamber.

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2	2		4.5.1.1.3.1 (a)
2			2	4.5.1.1.3.1 (b)
6	2	2	2	4.5.1.1.3.2 (a)

FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1 (c)
5	2	1	2	4.5.1.1.3.1 (d)

4.5.1.2.3 Dust.

4.5.1.2.3.1 Dust test.- This test shall be conducted in accordance with Method 510 of MIL-STD-810 except that the high temperature shall be 350°F for SMDC or 200°F for FCDC. After completion of this test, the specimens shall be test fired as follows:

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SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
2		2		4.5.1.1.3.1 (a)
4	2		2	4.5.1.1.3.1 (b)
3	1	1	1	4.5.1.1.3.2 (a)

FCDC

Total Units	Temperature in Accordance with 4.5.1.2.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1 (d)

4.5.1.2.4 Vibration.

4.5.1.2.4.1 Vibration test.- The test shall be conducted in accordance with Method 514 (aircraft category, procedure I, parts 1, 2 & 3 and curve H) of MIL-STD-810, except that for each resonant and cycling period the assemblies shall be divided equally for vibration at -65°F, & 70 and +350°F for SMCD & 200°F for FCDC. After vibration testing has been completed the assemblies shall be x-ray inspected and then test fired as follows:

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
2			2	4.5.1.1.3.1 (a)
4	2	2		4.5.1.1.3.1 (b)
3	1	1	1	4.5.1.1.3.2 (a)

FCDC

Test Firing - Upon completion of radiographic inspection, nine FCDC's shall be fired as follows:

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Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1 (c)
5	2	1	2	4.5.1.1.3.1 (d)

4.5.1.2.5 Shock.

4.5.1.2.5.1 Shock test.- This test shall be conducted in accordance with Method 516, procedure I of MIL-STD-810. The shock pulse waveform shall be a terminal peak saw tooth. The peak amplitude shall be 20g and the duration shall be 11 ms. Upon completion of shock testing, the assemblies shall be x-ray inspected, divided into three groups as follows and test fired at -65°F, +70°F, +350°F for SMDC and 200°F for FCDC.

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
2		2		4.5.1.1.3.1(a)
4	2		2	4.5.1.1.3.1(b)
3	1	1	1	4.5.1.1.3.2(a)

FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
6	2	2	2	4.5.1.1.3.1(c)
6	2	2	2	4.5.1.1.3.1(d)

4.5.1.2.6 Tensile test.- The 3 SMDC assemblies selected prior to forming shall be subjected to a tensile test. The end fitting joint between the outer ferrule and the stainless steel tubing shall withstand a minimum 200 lb. load at 4 cycles minute for 600 cycles. The assemblies shall then be subjected to a leak detection test per 4.4.3, and X-ray inspected per 4.4.5.

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Five FCDC shall be selected for a tensile test. The end fitting joint between the large ferrule and the outer braiding shall withstand a 300 pound minimum load. Upon completion of the tensile test the assemblies shall be test fired as follows:

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
3	1	1	1	4.5.1.1.3.3

FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
5	2	1	2	4.5.1.1.3.1(d)

4.5.1.2.7 Submerged operation test.- The 3 SMDC assemblies shall be submerged in water and then pressurized to 50 psig. After a period of 10 minutes at this environment the SMDC shall be fired as shown in Figure 6 at ambient temperature.

The three assemblies FCDC shall be submerged in water and pressurized to 50 psig. After a period of 10 minutes at this environment, three FCDC's shall be fired while submerged at ambient temperature. Firing shall be in accordance with 4.5.1.1.3.1(d).

4.5.1.2.8 Temperature.

4.5.1.2.8.1 High temperature test.- This test shall be conducted in accordance with Method 501, procedure I of MIL-STD-810 except that the SMDC's shall be exposed to a temperature of +350°F and FCDC's to 200°F until thermal stabilization is reached and shall then be maintained at that temperature for a period of 50 hours. Then the assemblies shall be test fired as follows:

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2	2		4.5.1.1.3.1 (a)
2			2	4.5.1.1.3.1 (b)
6	2	2	2	4.5.1.1.3.2 (a)

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FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1 (c)
5	2	1	2	4.5.1.1.3.1 (d)

4.5.1.2.8.2 Low temperature test.- Six assemblies shall be placed within the test chamber and the chamber cooled to and maintained at a temperature of -65°F for 72 hours minimum after stabilization of the unit temperature is reached. Upon completion of the low temperature test half of the assemblies shall be test fired within five minutes after removal from the test chamber. The remaining half shall be X-ray inspected and then test fired.

4.5.1.2.9 Auto detonation test.- Two SMDCs shall be placed in an oven preheated to 400°F for one hour and two FCDCs to 350°F for one hour. Aluminum witness plates affixed in intimate contact with the end of the boosters will record whether or not the SMDCs and FCDCs auto detonate. A 0.005 inch dent in the witness plate is evidence of auto detonation. If no SMDCs or FCDCs auto detonate during one hour, the test is complete. If high order detonation occurs, testing shall be terminated and the reason(s) for the failure determined and corrected.

4.5.1.2.10 High temperature exposure (12 hours).- Two SMDCs shall be placed in an oven preheated to a temperature of +375°F maintained for 12 hours. If no SMDCs detonates, the two SMDCs shall be cooled to 70°F and functionally tested. If any SMDCs detonates, or fails to meet design performance requirements after cooling, the test shall be repeated with additional groups of SMDCs, decreasing the temperature in increments of 25°F, until design performance requirements are met.

Two FCDC's shall be placed in an oven preheated to a temperature 325°F for exposure without detonation in 4. The temperature shall be maintained for 12 hours. If no FCDC detonates the two FCDC shall be cooled to 70°F and functionally tested. If any FCDC detonates or fails to meet design performance requirements after cooling, the test shall be repeated with additional groups of FCDC, decreasing the temperature in increments of 25°F, until design performance requirements are met.

SMDC

Total Units	Temperature in Accordance with 4.5.1.1.4	Test Fire in Accordance with Paragraph
	Ambient	
2	2	4.5.1.1.3.1(a)
2	2	4.5.1.1.3.1(b)
2	2	4.5.1.1.3.2(a)

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FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4	Test Fire in Accordance with Paragraph
	Ambient	
8	8	4.5.1.1.3.1(c)
4	4	4.5.1.1.3.1(d)

4.5.1.3 Minimum bend test. Six SMDC assemblies shall be formed in 3/4 inch radius helix along the centerline for two complete revolutions with adjacent coils in intimate contact. The six SMDCs shall then be test fired as follows:

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
2		2		4.5.1.1.3.1(a)
4	2		2	4.5.1.1.3.1(b)

4.5.1.4 Forty-foot drop test.- The six assemblies shall be conducted in accordance with test 103 of MIL-STD-331. Two assemblies shall be dropped in the longitudinal axis nose up position, two in the longitudinal axis nose down position and two in the longitudinal axis horizontal position. The assemblies shall be dropped with the end boosters unprotected and no detonations shall occur.

4.5.1.5 Six-foot drop test.- The assemblies with the end boosters protected by shipping covers, shall be dropped from a height of six feet onto a minimum thickness of one foot of reinforced concrete. The assemblies shall not detonate if they are bent the assemblies shall be pressed to the original configuration and shall be capable of satisfactory operation after completion of this test. The drop axis shall be as follows:

(a) One drop on each end with the longitudinal axis perpendicular to the slab.

(b) One drop with longitudinal axis parallel to the slab.

Upon completion of the drop test, the assemblies shall be X-ray inspected and then test fired as follows:

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Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
4	2		2	4.5.1.1.3.1(a)
2		2		4.5.1.1.3.1(b)

FCDC

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
6	2	2	2	4.5.1.1.3.1(d)

4.5.1.6 Flexibility test.- Nine FCDCs shall be held firmly by one end on a surface, flexed 20° in one direction, brought back to the vertical position, then flexed 20° in the opposite direction. This operation shall be repeated for 20,000 flexures. (See 2 on Table II). The nine FCDCs shall then be formed with adjacent coils in contact around a two and one half inch diameter mandril and test fired as follows:

Total Units	Temperature in Accordance with 4.5.1.1.4			Test Fire in Accordance with Paragraph
	Low	Ambient	High	
9	3	3	3	4.5.1.1.31(e)

4.5.2 Preproduction test report.- The contractor shall-submit to the procuring activity, for approval, copies of his preproduction test report. The report shall use the format of MIL-STD-831 and include the test results, all supporting data collected in conducting the tests, and analysis of all failures which occurred. As a supplement the contractor may include, for the procuring activity's consideration, any recommendations for design improvement beyond any corrective action required to pass the test. The data presentation of the test report shall be coordinated with the approved test procedure required by 4.7.

4.5.3 Preproduction approval.- Satisfactory completion of the preproduction test requirements shall be contingent upon the procuring activity's approval of the contractor preproduction test report. Fabrication of production devices prior to such approval is at the contractor's own risk. After completion of the tests, the disposition of the test samples tested shall be as specified in the contract.

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4.6 Special tests.- Changes in design and material subsequent to completion of the preproduction tests may require rework and retesting of the devices involved. Such tests as directed by the procuring activity for the purpose of evaluating the changes on the performance and quality of the revised SMDCs and FCDCs are classified as Special Tests.

4.6.1 Scope of special tests.- Test procedures previously approved by the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.7 Test plans and procedures.- Test procedures shall be in accordance with the applicable requirements of Specification MIL-T-18303. The plans and procedures used for conducting Preproduction Tests shall be prepared by the contractor and submitted to the procuring activity for approval. The right is reserved by the procuring activity to modify the tests or to require any additional test procedures or details deemed necessary to determine compliance with the requirements of this specification or the contract. The test details, such as the test fixture design, the length of characteristics to be measured, special failure criteria, thermocouple and accelerometer locations, instrumentation description, etc., shall be part of the test procedures to be submitted.

4.8 Rejection and retest.- The assemblies which have been rejected may be reworked, or have parts replaced to correct the defects, and resubmitted for acceptance upon approval of the procuring activity. Before resubmittal, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the procuring activity. After corrections have been made, all tests deemed necessary by the procuring activity shall be repeated. No device that has been fired may be reworked and resubmitted.

NOTE: The following action shall be taken in case of a failure:

(a) Immediately notify the procuring activity's representative.

(b) Determine the cause of failure.

(c) Determine if the failure is a recurring manufacturing problem or design deficiency.

(d) Submit analysis to the procuring activity.

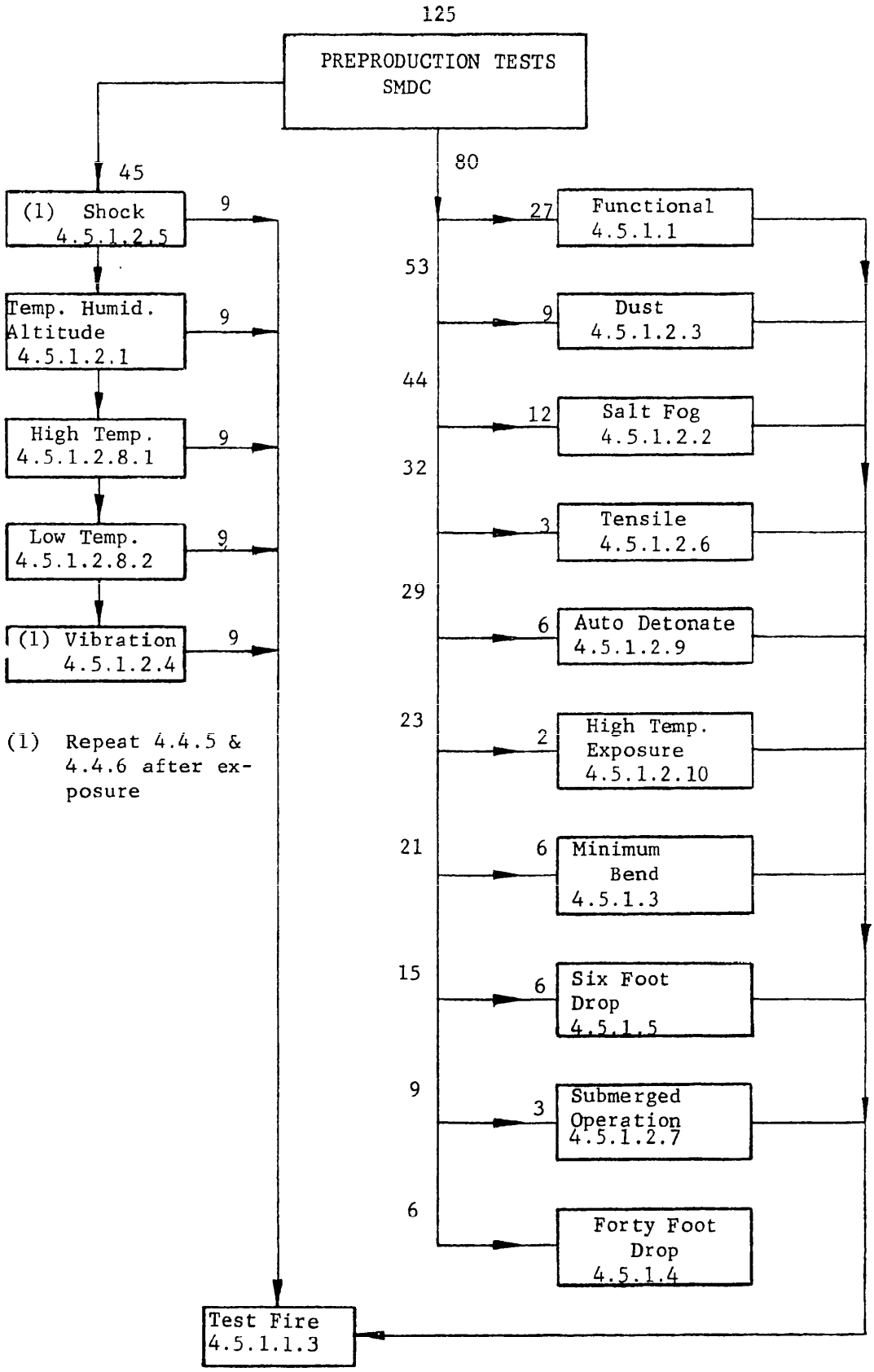
(e) Submit to the procuring activity for approval the proposed corrective action intended to reduce the possibility of the same failure(s) recurring.

5. PREPARATION FOR DELIVERY

5.1 General.- Each assembly shall be preserved, packaged, and packed for the level of shipment specified in the contract or purchase order in accordance

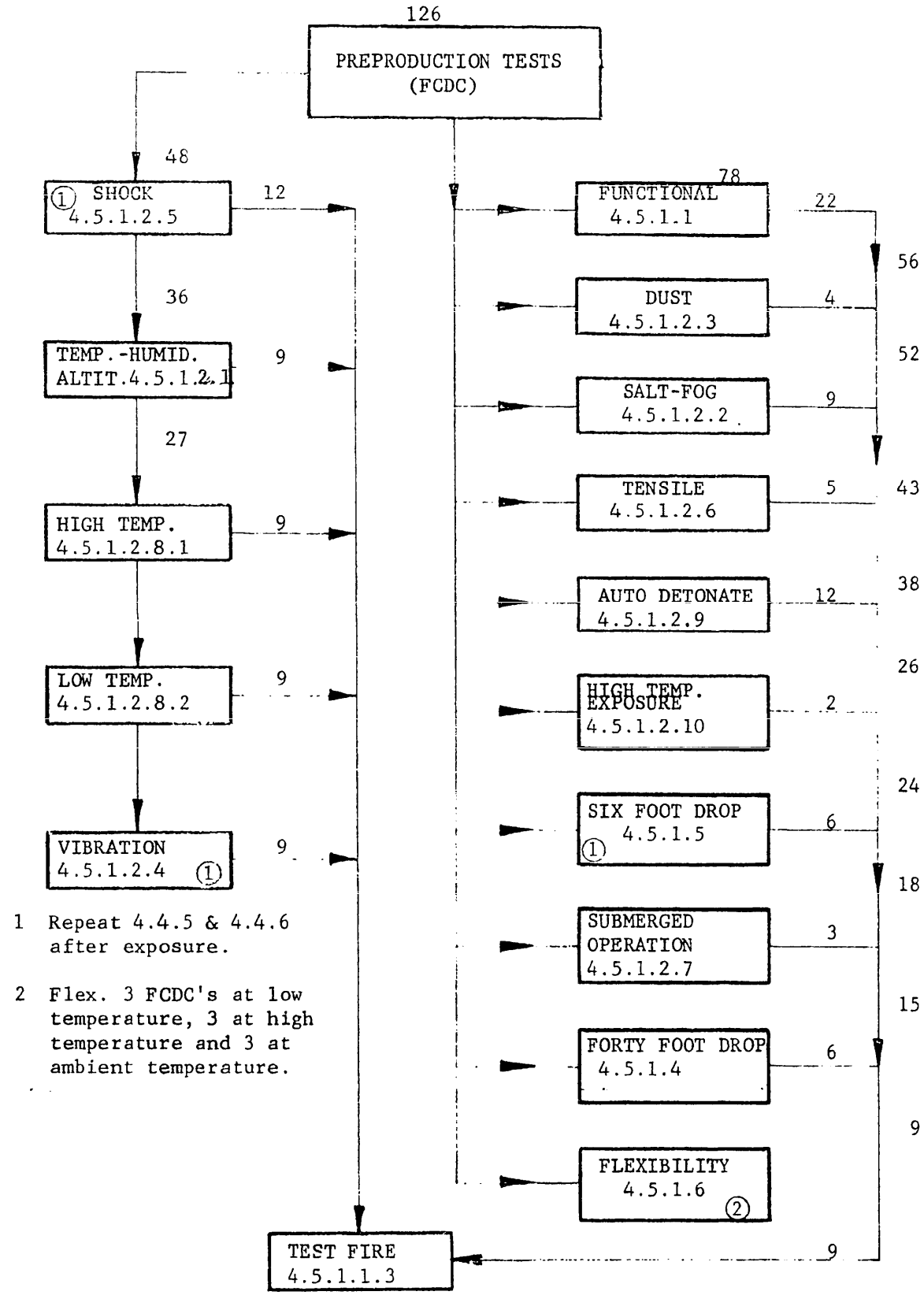
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TABLE I - PREPRODUCTION TEST SEQUENCE



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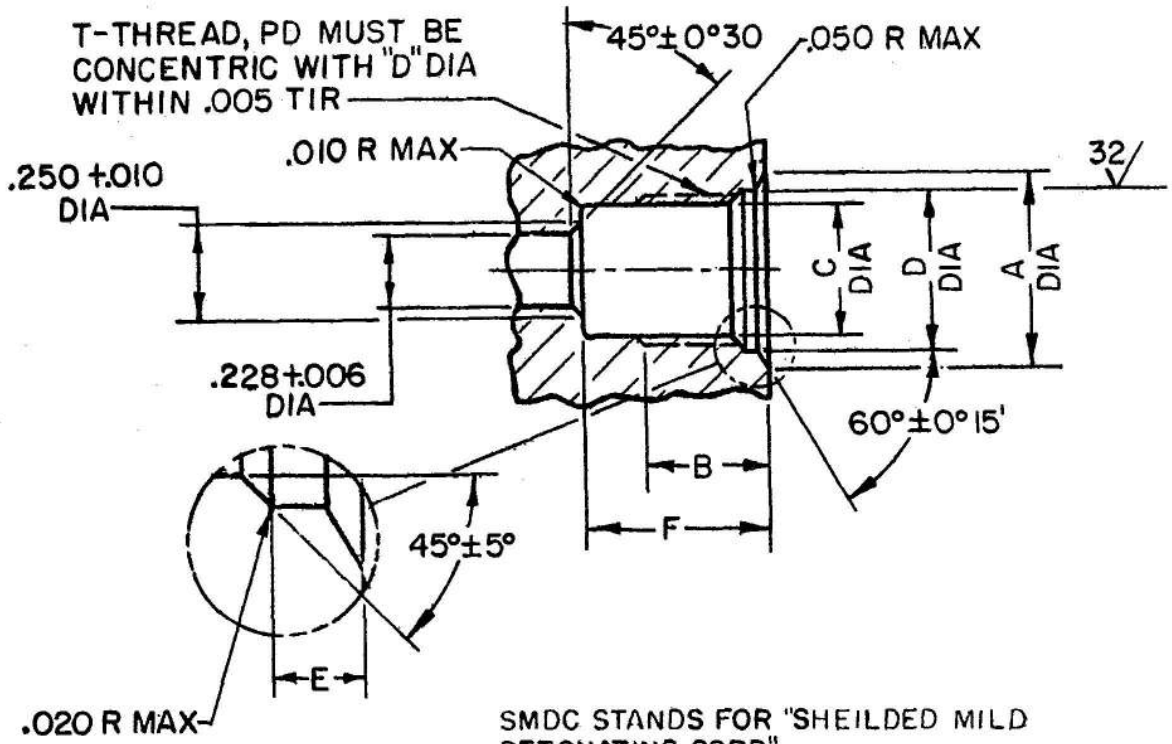
TABLE II - PREPRODUCTION TEST SEQUENCE



1 Repeat 4.4.5 & 4.4.6 after exposure.

2 Flex. 3 FCDC's at low temperature, 3 at high temperature and 3 at ambient temperature.

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SMDC STANDS FOR "SHEILDED MILD DETONATING CORD"

T THREAD PER MIL-S-7742	A DIA +.015	B MIN THD DP	C DIA +.002	D DIA +.005	E +.015	F -.002
.3750-24UNF-3B	.500	.325	.330	.390	.063	.473
.4375-20UNF-3B	.562	.400	.385	.454	.075	.532
.5000-20UNF-3B	.625	.400	.453	.517	.075	.532

REQUIREMENTS

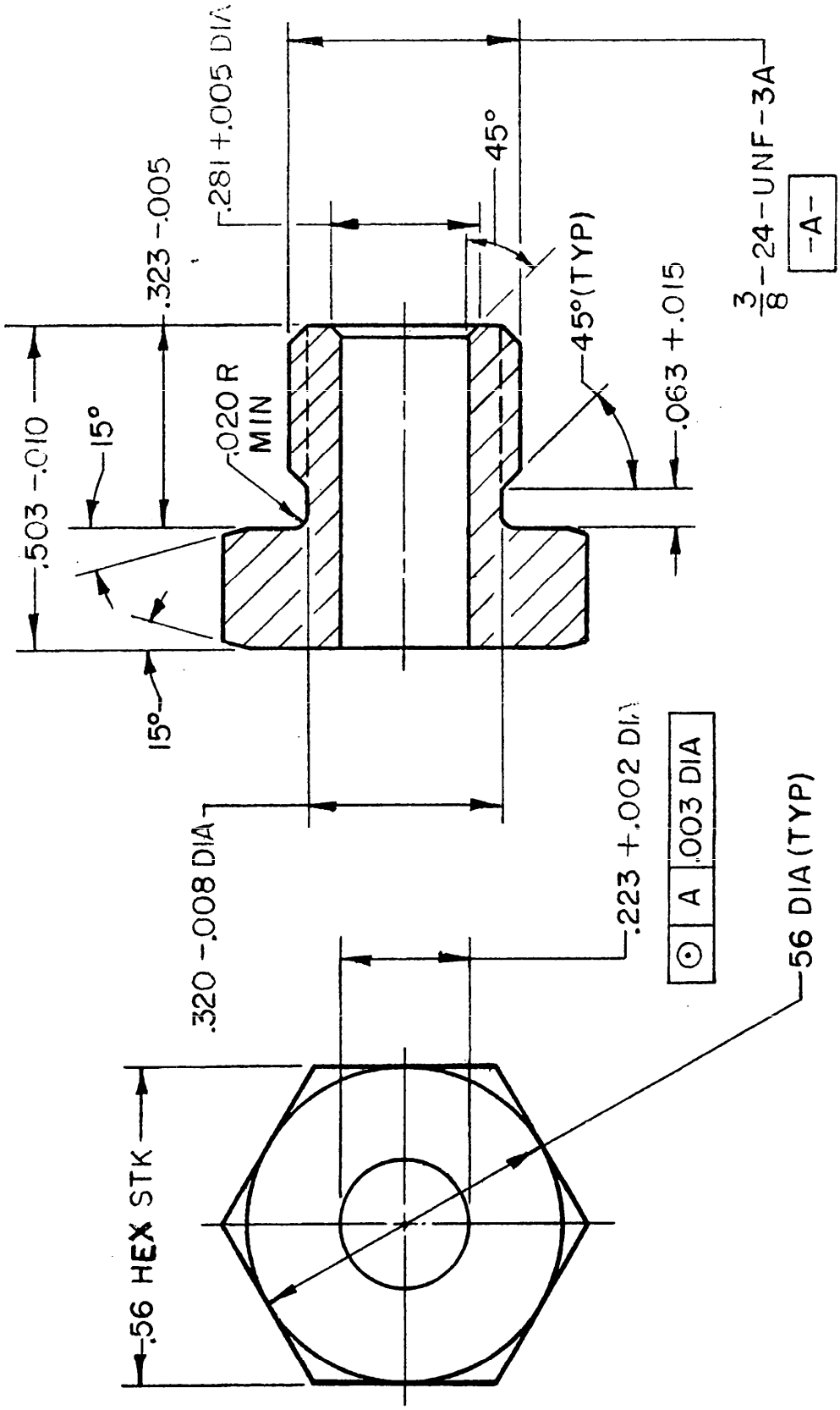
1. GENERAL
   - 1.1. BOSSES SHALL MEET THE REQUIREMENTS SPECIFIED HEREIN
   - 1.2. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER THE DOCUMENTS REFERENCED HEREIN WHICH SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID
2. DIMENSIONS IN INCHES, TOLERANCES: AS SHOWN
3. SURFACE ROUGHNESS: 125 MICROINCHES PER USAS B46.1, EXCEPT AS NOTED
4. REMOVE ALL BURRS AND SLIVERS, ALSO BREAK SHARP EDGES

FIGURE 1

BOSS

SMDC PORT CONFIGURATION

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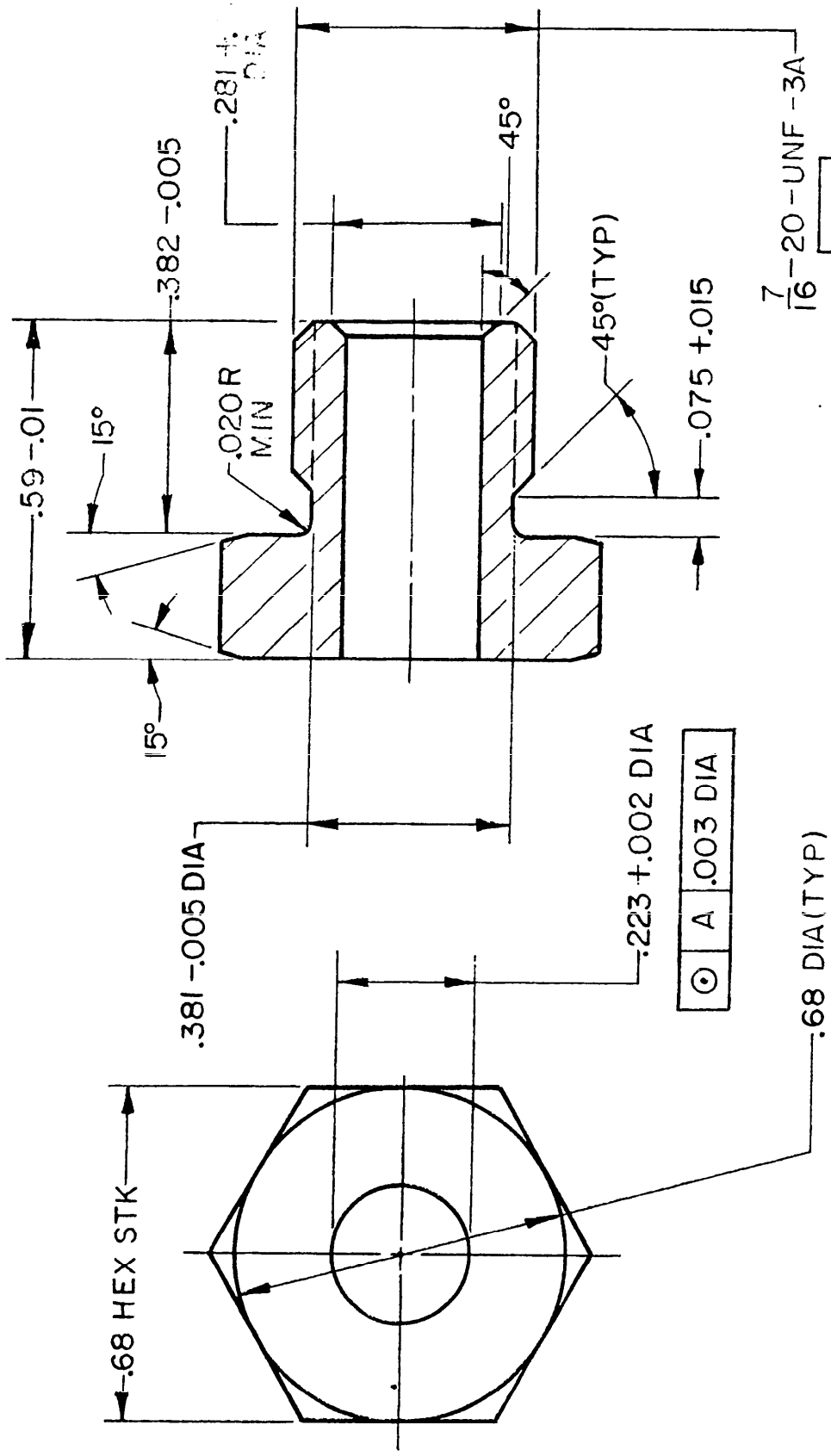
TOL ON ANGLES = 1°

NO SCALE

FITTING NUT

FIGURE 2A

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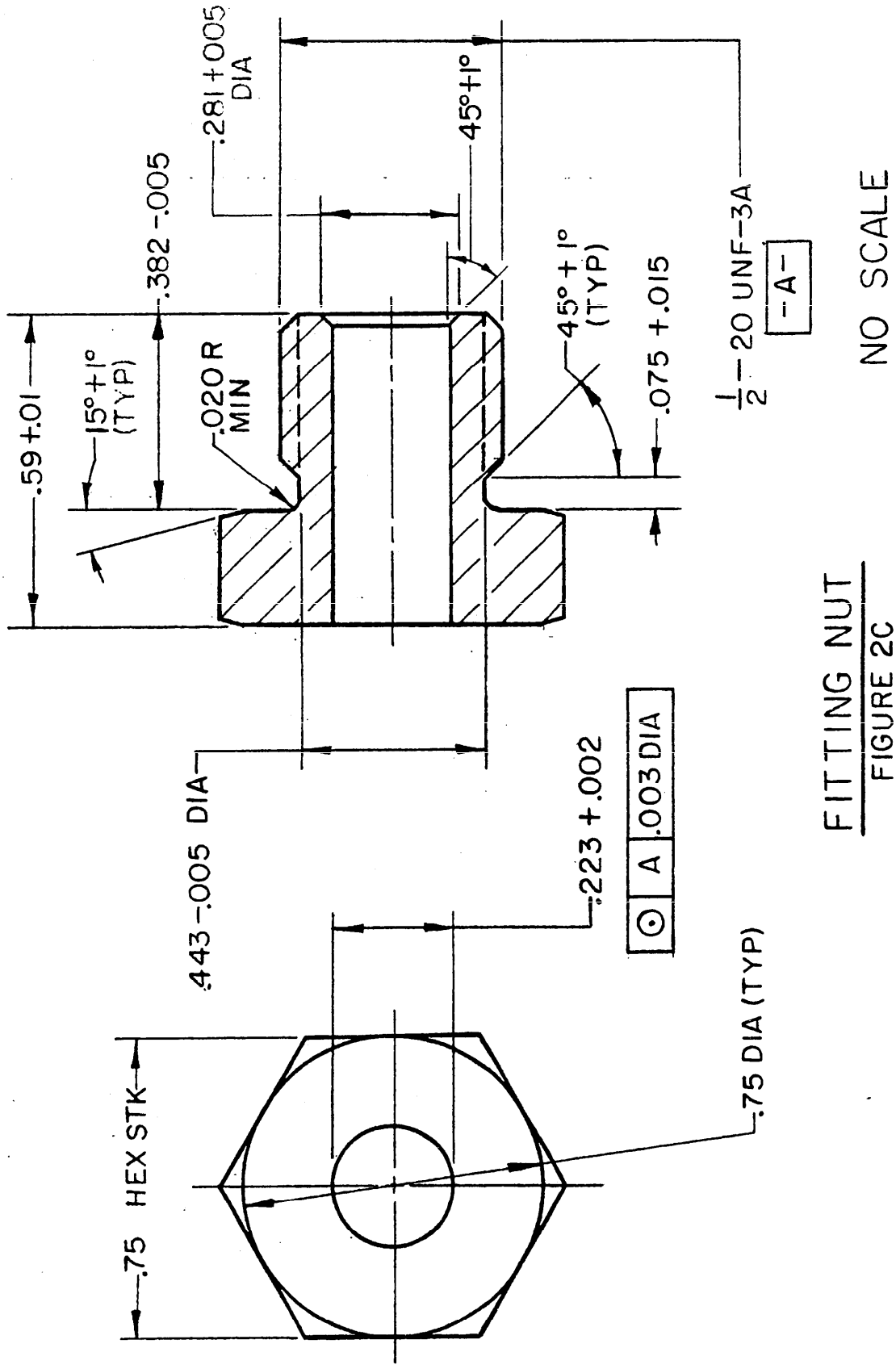
TOL ON ANGLES = 1°

NO SCALE

FITTING NUT

FIGURE 2B

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FITTING NUT
FIGURE 2C

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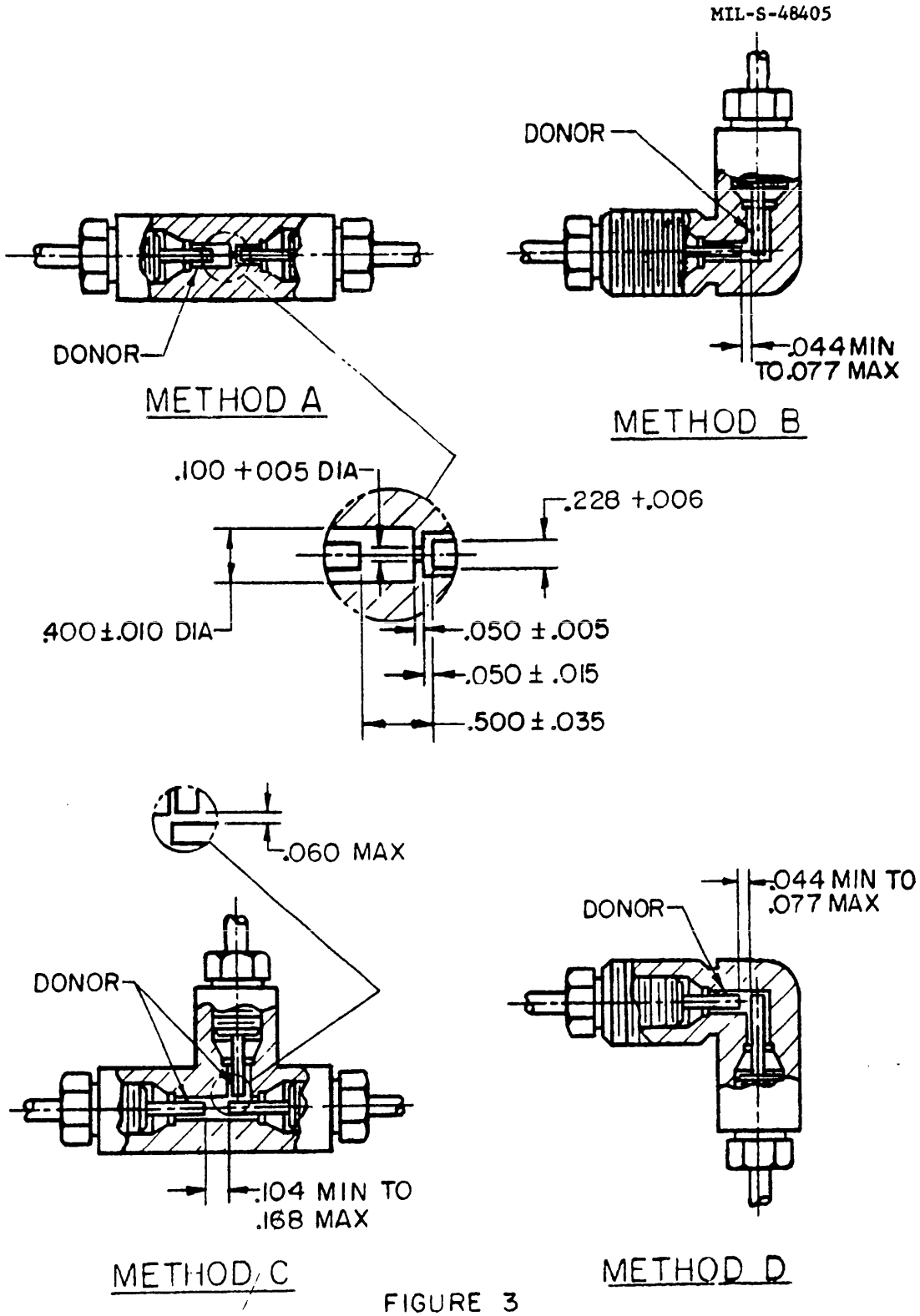
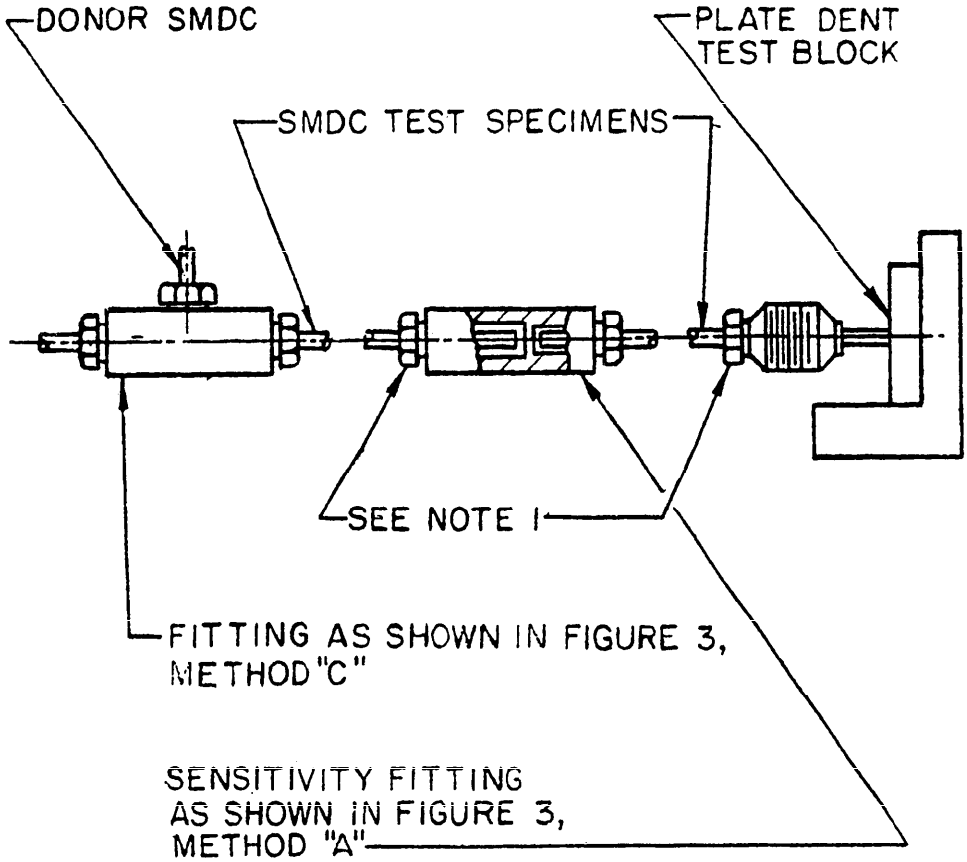


FIGURE 3
FIRING ALIGNMENTS AND GAPS

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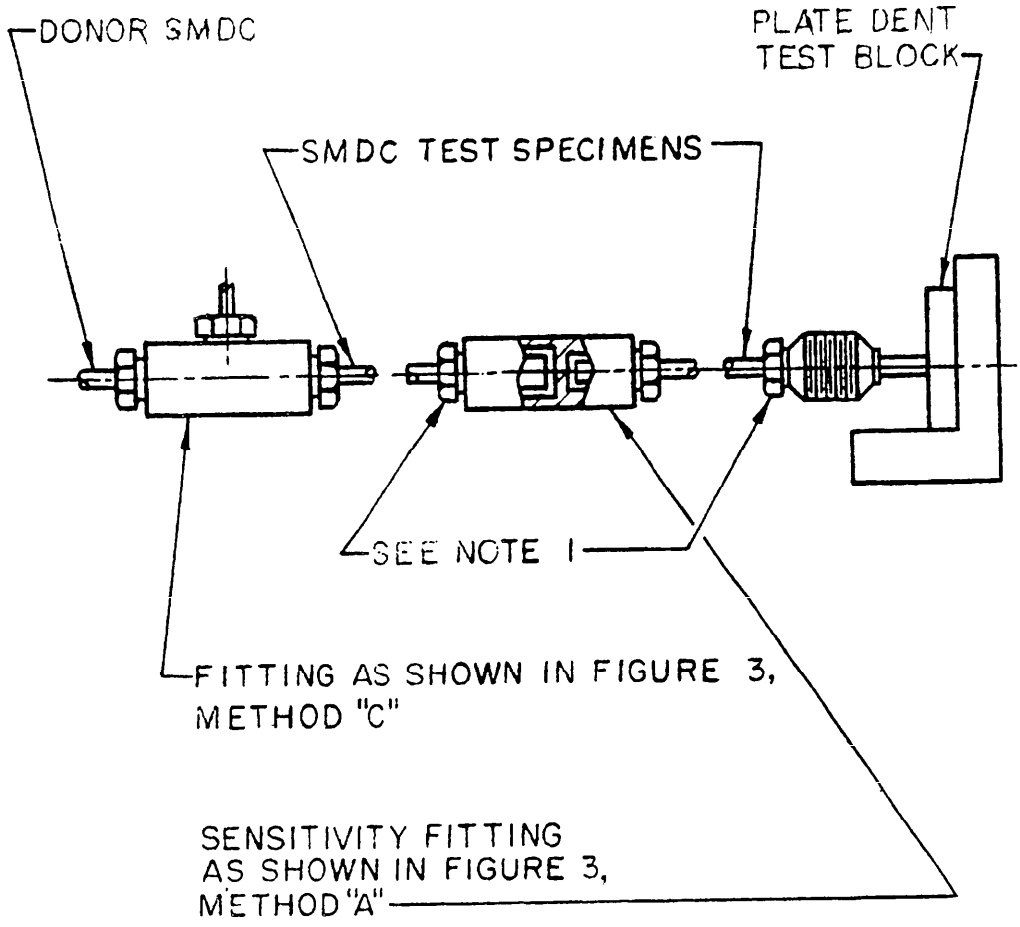


NOTE 1 WHEN TESTING THE OVERLOAD SMDC ASSEMBLIES THIS END SHALL CONTAIN THE INCREASED CHARGE

FIGURE 4

FIRING SCHEMATIC

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NOTE 1 WHEN TESTING THE OVERLOAD SMDC ASSEMBLIES THIS END SHALL CONTAIN THE INCREASED CHARGE

FIGURE 5

FIRING SCHEMATIC

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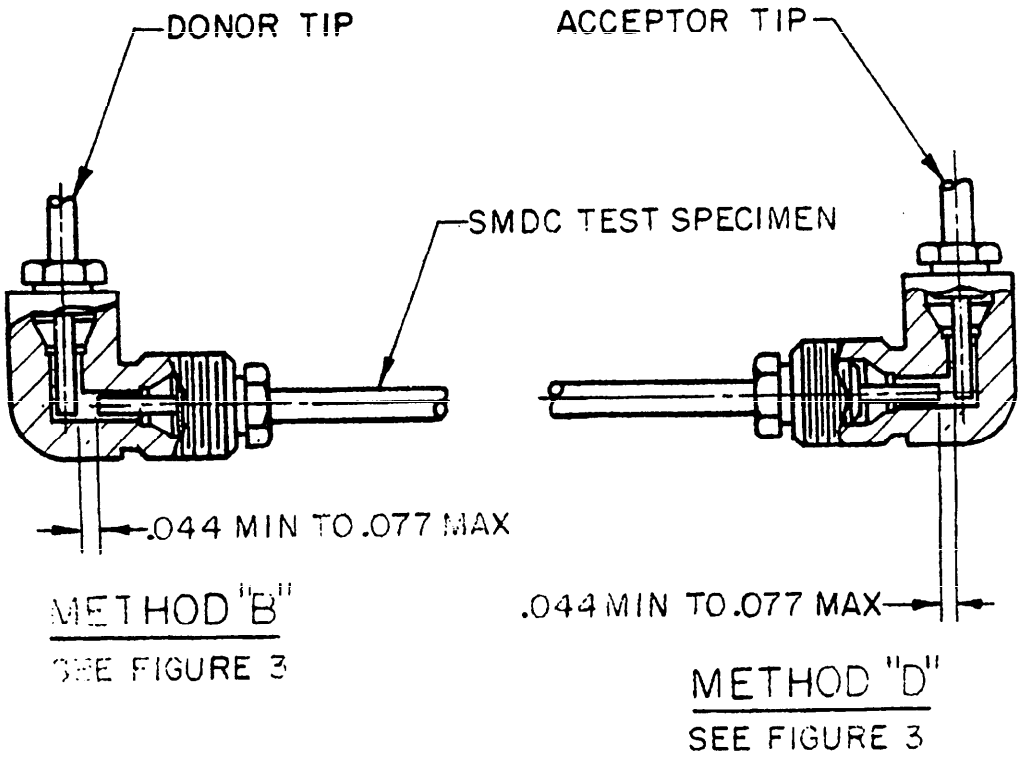


FIGURE 6
FIRING SCHEMATIC

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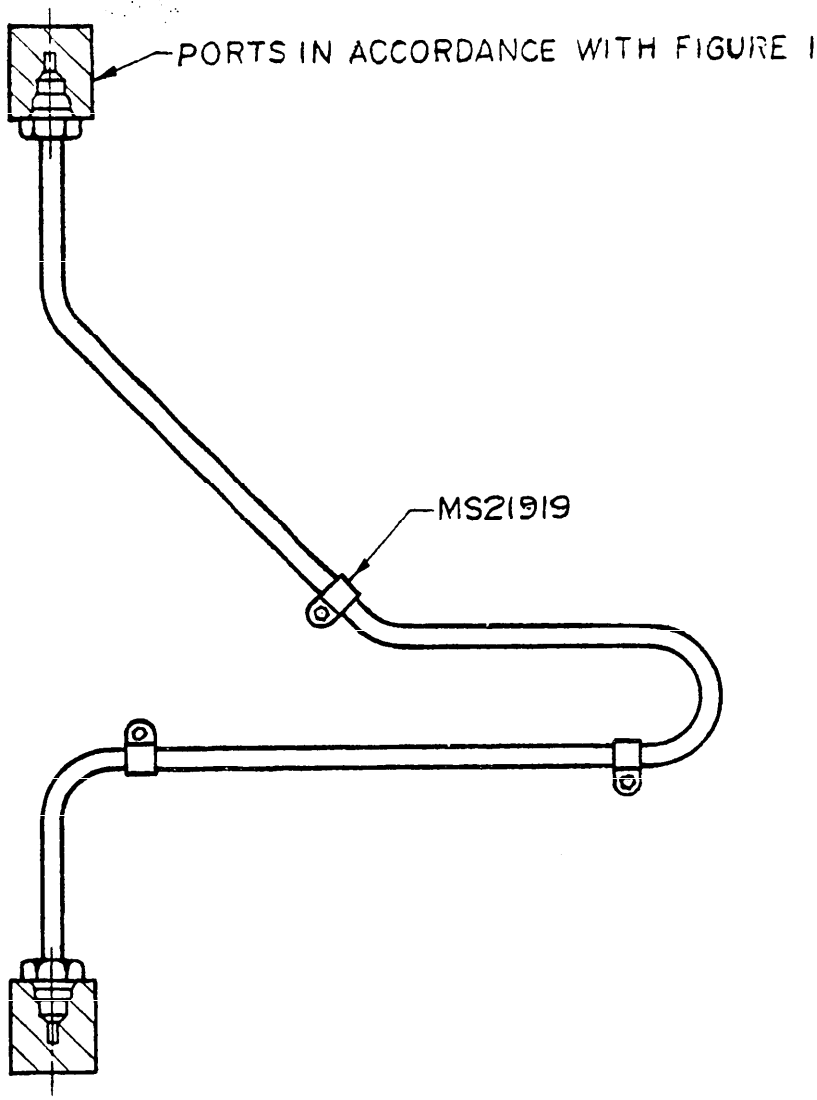


FIGURE 7

VIBRATION TEST SETUP

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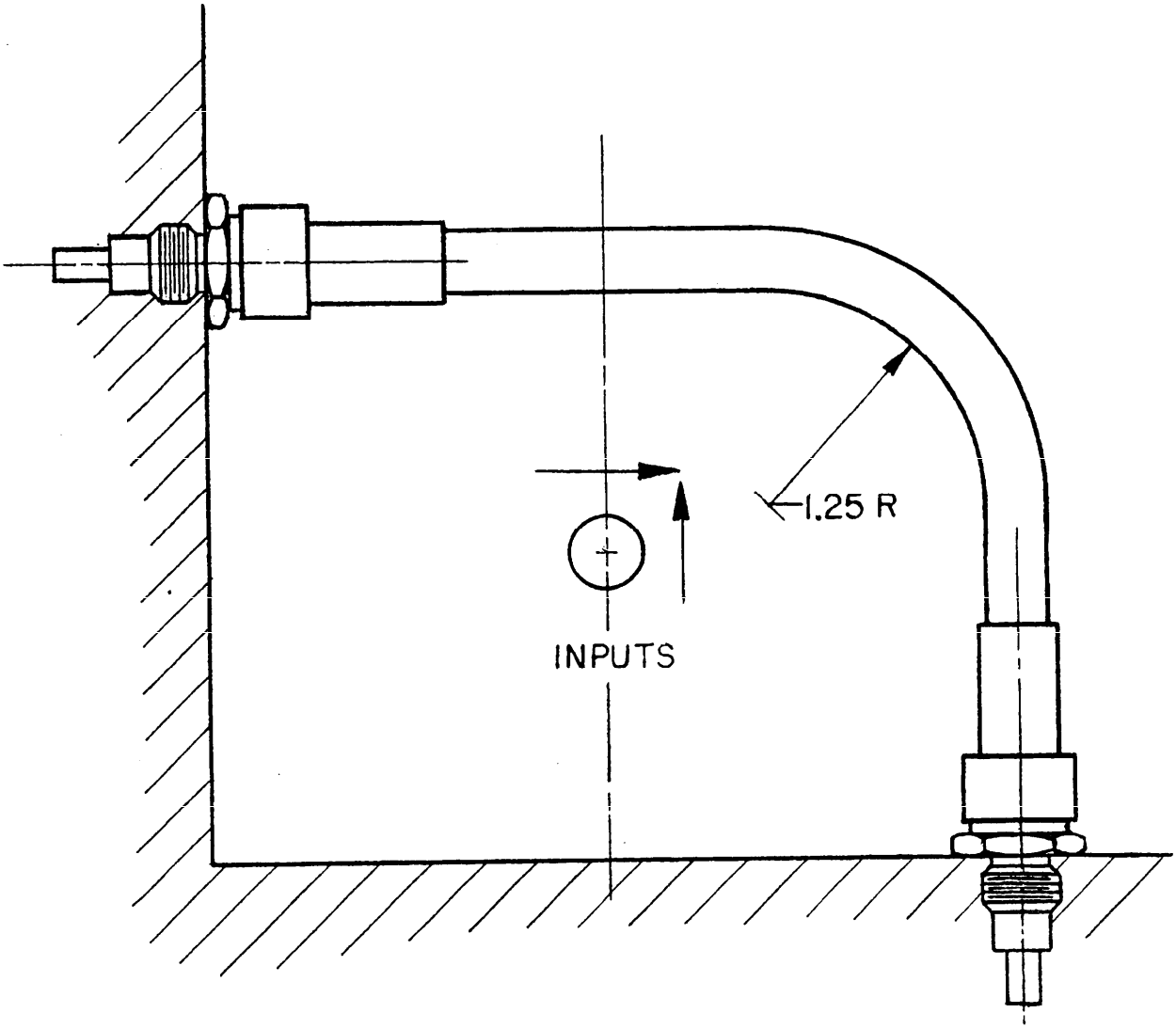
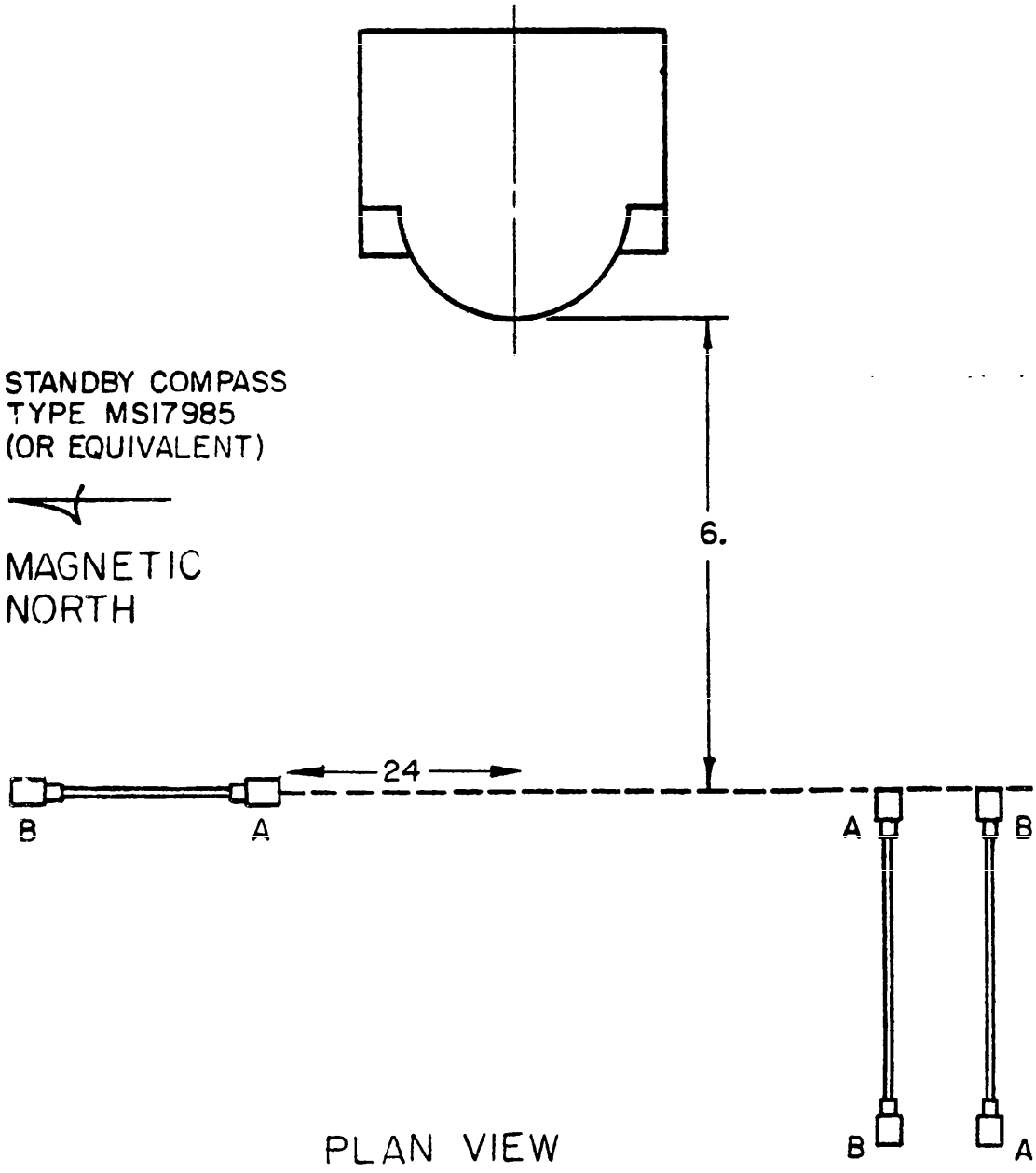


FIGURE 8

VIBRATION MOUNT CONCEPT

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NOTE 1. ASSEMBLY AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 9

LAYOUT FOR DETERMINING OF RISIDUAL MAGNETISM IN ASSEMBLY

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with MIL-STD-794.

6. NOTES

6.1 Intended use.- The assemblies covered by this specification are intended for use in the Emergency Escape System of Aircraft.

6.2 Precedence of documents.- When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This specification - This specification and its applicable drawing shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications.- Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance objectives.- Minimum size and weight, simplicity of operation, case of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment.

6.4 Testing ground rules.

6.4.1 General rules.- The following ground rules apply to the design verification, preproduction, acceptance and special tests required herein, and supplement those found in the referenced specifications.

(a) A survey shall be conducted on one assembly prior to test to determine the time required to reach thermal stability. This shall be accomplished by monitoring a thermocouple located as closely as possible to the core of the item of greatest thermal inertia.

(b) Throughout the program and after every design change the assembly shall be tested at a unit level and in addition the assembly's and the test fixture(s) shall be tested as an integrated system to assure compatibility.

6.5 Safety procedures.- Standard DOD 4145.26M may be used as a guide.

Custodian:

Army - MU
Air Force - 11

Preparing activity:

Army - MU

Project No. 1377-0565

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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

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